



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE(Advanced)
FULL SYLLABUS
26-03-2023

JEE(Main + Advanced) : ENTHUSIAST COURSE : [SCORE-II (ALL STAR BATCH)]

Time : 3 Hours

Maximum Marks : 180

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **40** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140,

- | | |
|------------------------------------|--|
| • Boltzmann constant | $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$ |
| • Coulomb's law constant | $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ |
| • Universal gravitational constant | $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$ |
| • Speed of light in vacuum | $c = 3 \times 10^8 \text{ ms}^{-1}$ |
| • Stefan-Boltzmann constant | $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$ |
| • Wien's displacement law constant | $b = 2.89 \times 10^{-3} \text{ m-K}$ |
| • Permeability of vacuum | $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$ |
| • Permittivity of vacuum | $\epsilon_0 = \frac{1}{\mu_0 c^2}$ |
| • Planck constant | $h = 6.63 \times 10^{-34} \text{ J-s}$ |

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I (i) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

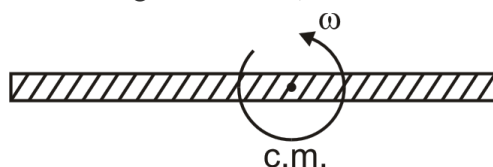
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

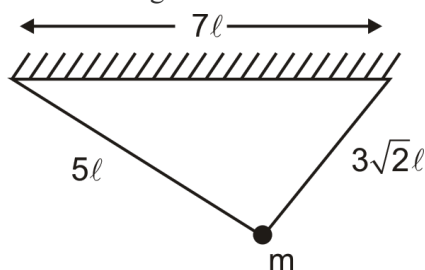
1. A thin rod has an unknown non-uniform mass density. It is mounted on an axle passing perpendicular to it, through its centre of mass as shown and is then rotated about the axle. The axle divides the rod into two parts, one on each side of it. Which of the following must be true, no matter how the mass in the rod is distributed ?



- (A) The two parts must have the same mass
- (B) The magnitude of the linear momentum of the two parts must be equal.
- (C) The magnitude of the angular momentum of the two parts about the centre of mass must be equal.
- (D) The kinetic energies of two parts must be equal.

Space for Rough Work

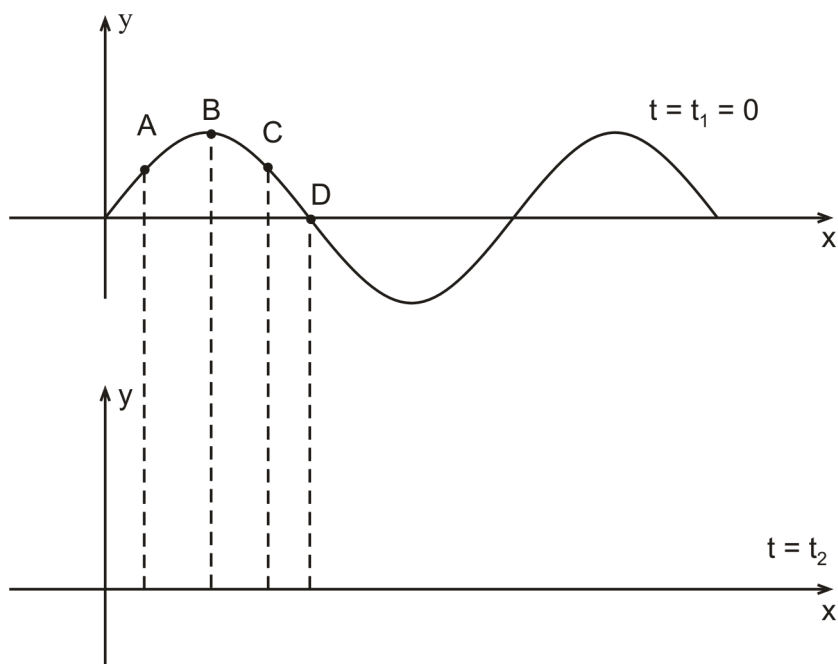
2. A particle is suspended by two ideal strings as shown in the figure. Now mass m is given a small displacement perpendicular to the plane of triangle formed. Choose the correct statement(s).



- (A) Period of oscillation of system is $2\pi\sqrt{\frac{3\sqrt{3}l}{g}}$
- (B) The period of oscillation of system is $2\pi\sqrt{\frac{3l}{g}}$
- (C) The period of oscillation of system is independent of m .
- (D) If the distance between the suspension points was kept constant and the length of the strings were quadrupled, then the period of the system will get doubled.
3. Ball A is in motion with linear momentum P while ball B is at rest. Mass of both the balls is same & equal to m . Ball A obliquely collides with ball B. After the collision, A moves at an angle 30° with the initial direction while momentum of ball B is $\frac{2}{\sqrt{3}}$ times of its minimum value. Then which option(s) is/are correct.
- (A) After the collision both ball A & B move at angle of 60° with each other.
- (B) After the collision both ball A & B move at angle of 90° with each other.
- (C) After the collision, momentum of ball A is $\frac{P}{\sqrt{3}}$
- (D) Loss in K.E. of system after the collision is $\frac{P^2}{6m}$.

Space for Rough Work

4.

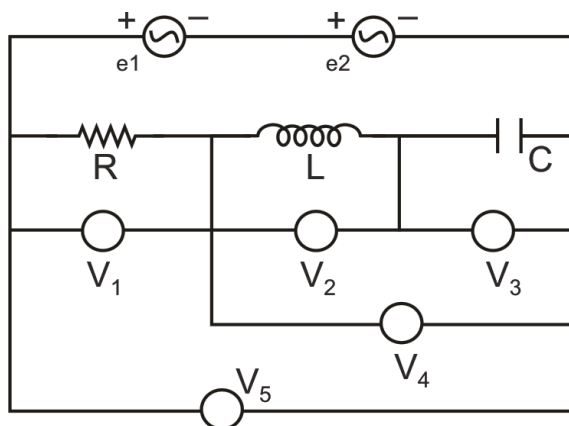


The graph shows the standing wave on a string at two successive instants of time t_1 & t_2 . A, B, C, D are the points on the string. (B is the point corresponding to maximum displacement amplitude of standing wave). Select the correct option(s) regarding observations about net mechanical energy for the interval between t_1 & t_2 .

- (A) For A, Energy is moving towards right through this point.
- (B) For B, no net energy ever crosses this point.
- (C) For C, Energy is moving towards left through this point.
- (D) For A, Energy is moving towards left through this point.

Space for Rough Work

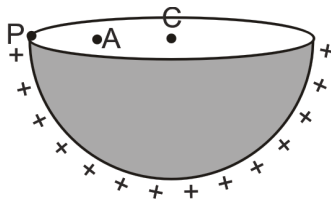
5. A resistor $R = 500 \, \Omega$, an ideal inductor $L = 0.9 \, \text{H}$ and a capacitor $C = 2.5 \, \mu\text{F}$ are connected with two alternating source $e_1 = 100 \sin(1000t)$ and $e_2 = 100\sqrt{3} \cos(1000t)$. V_1, V_2, V_3, V_4 and V_5 are five ideal voltmeters and connected in the circuit as shown in the figure. Choose the correct option(s).



- (A) Phase difference between voltages of V_3 and V_5 is $\frac{\pi}{2} + \tan^{-1} \left(\frac{9}{5} \right)$
- (B) Reading of V_4 is 180 volt.
- (C) Difference between reading of V_2 and V_1 is 80 volt.
- (D) Ratio of reading of V_4 and V_5 is $\frac{1}{\sqrt{2}}$

Space for Rough Work

6. A non conducting hemispherical shell of radius 'R' has uniform surface charge density σ . Point 'C' is center of hemispherical shell. Point A and point 'P' lie on the flat surface of the shell. The distance between C and A is $R/2$ and between C and P is R. Choose the correct option(s).



- (A) The potential at point P is $\frac{\sigma R}{2\epsilon_0}$
- (B) The potential at point A is $\frac{\sigma R}{4\epsilon_0}$
- (C) Work done by electrostatic force if a charge ' q_0 ' is moved from P to A is $\frac{-q_0 \sigma R}{4\epsilon_0}$
- (D) Work done by electrostatic force if a charge ' q_0 ' is moved from P to A is $\frac{q_0 \sigma R}{4\epsilon_0}$

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 12)

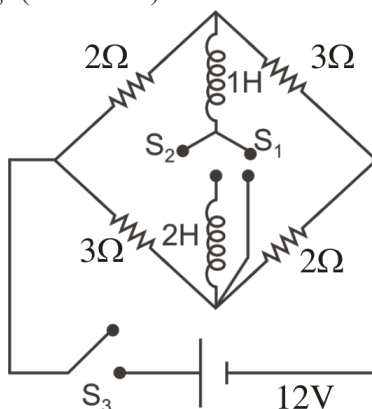
- This section contains **FOUR (04)** questions.
- Each **question has matching lists**. The codes for the lists have choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If ONLY the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

7. Four resistors and two ideal inductors $1H$ and $2H$ and a battery of $12V$ are arranged in given circuit with the help of three switches S_1 , S_2 & S_3 . ($e^{-1} = 0.37$)



List-I, describes the statement regarding the given circuit and List-II gives their values (either exactly or as a close approximation).

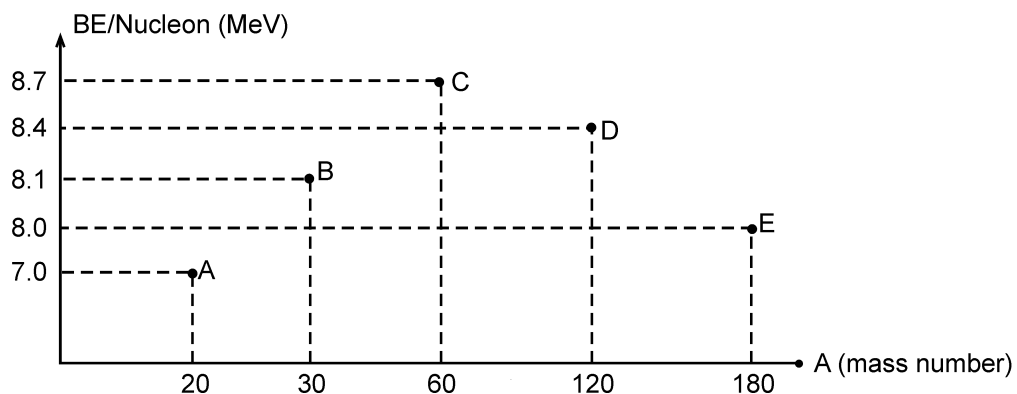
List-I		List-II	
(I)	When switch S_1 and S_3 are closed simultaneously and S_2 remain opened, then time constant of circuit in seconds.	(P)	1
(II)	When switch S_1 and S_3 remain closed for long time and S_2 remain opened then current through inductor in ampere.	(Q)	$5/12$
(III)	When S_1 and S_3 remain closed for long time and S_2 remain opened then current in battery.	(R)	$1/3$
(IV)	If S_1 and S_3 remain closed for long time and then S_1 is opened S_2 is closed and S_3 remain closed then current (in A) in the branch of inductor immediately after switching	(S)	5
		(T)	$2/3$

Which one of the following options is correct.

- (A) $I \rightarrow P; II \rightarrow Q; III \rightarrow T; IV \rightarrow S$ (B) $I \rightarrow Q; II \rightarrow P; III \rightarrow T; IV \rightarrow S$
 (C) $I \rightarrow Q; II \rightarrow P; III \rightarrow S; IV \rightarrow R$ (D) $I \rightarrow Q; II \rightarrow P; III \rightarrow R; IV \rightarrow T$

Space for Rough Work

8.



Consider the position of different nuclei on the binding energy per nucleon graph as shown.

List-I (Reaction)		List-II (Q - Value (MeV))	
(I)	$2B \rightarrow C$	(P)	zero
(II)	$2B + C \rightarrow D$	(Q)	102
(III)	$E \rightarrow C + D$	(R)	non-zero
(IV)	$3A \rightarrow C$	(S)	36
		(T)	90

Which of the following options is correct ?

- (A) $I \rightarrow P, S; II \rightarrow P; III \rightarrow R, T; IV \rightarrow Q, R$ (B) $I \rightarrow R, S; II \rightarrow P; III \rightarrow P, T; IV \rightarrow Q, R$
 (C) $I \rightarrow S, T; II \rightarrow P, R; III \rightarrow R, T; IV \rightarrow Q, R$ (D) $I \rightarrow R, S; II \rightarrow P; III \rightarrow R, T; IV \rightarrow Q, R$

Space for Rough Work

9. A uniform candle of length 10 cm is glued with an aluminium cylinder of same radius and length 1 cm. Composite system is submerged in the water and it floats stably in vertical orientation. Density of wax is 0.8 g/cm^3 , density of aluminium is 2.7 g/cm^3 and density of water is 1.0 g/cm^3 . At $t = 0$ candle is lit and assume that it consumes wax at a constant rate of 0.1 cm/min . List – I gives four different instant of time while List –II lists magnitude of time period :

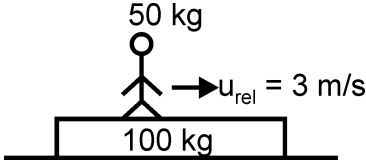
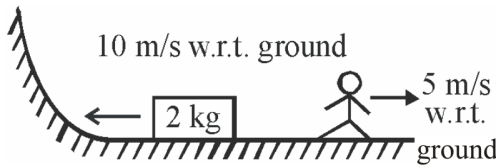
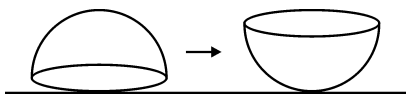
List-I		List-II	
(I)	$t = 5 \text{ minutes}$	(P)	$\sqrt{\frac{103}{107}}$
(II)	$t = 10 \text{ minutes}$	(Q)	$\sqrt{\frac{101}{107}}$
(III)	$t = 15 \text{ minutes}$	(R)	$\sqrt{\frac{91}{107}}$
(IV)	$t = 20 \text{ minutes}$	(S)	$\sqrt{\frac{99}{107}}$
		(T)	$\sqrt{\frac{95}{107}}$

The correct match for time period of small oscillation in each case will be
(in units of T_0 where T_0 is time period at $t = 0$)

- (A) $I \rightarrow P; II \rightarrow S; III \rightarrow T; IV \rightarrow T$ (B) $I \rightarrow P; II \rightarrow P; III \rightarrow Q; IV \rightarrow R$
(C) $I \rightarrow Q; II \rightarrow R; III \rightarrow P; IV \rightarrow Q$ (D) $I \rightarrow P; II \rightarrow R; III \rightarrow Q; IV \rightarrow Q$

Space for Rough Work

10. List-I describes for systems and List-II gives possible value of work done in Joules.

List-I		List-II	
(I)	A plank of mass 100 kg is placed on a smooth horizontal surface at rest along with a man. Now man starts moving on the plank with velocity of 3 m/s with respect to plank. The work done by man is. 	(P)	+ 100 J
(II)	A block of mass 2 kg is placed on a smooth track and imparted a velocity of 10 m/s to the left and observed by an observer moving with 5 m/s as shown in figure. Work done by the normal force applied by the track on the block up to the instant when the block reaches the highest point on the track in the frame of observer. Assume that block never loose contact with the track ($g = 10 \text{ m/s}^2$) 	(Q)	- 100 J
(III)	A particle of mass 1 kg is projected from an unstable equilibrium point N towards a stable equilibrium point S with negligible velocity in a conservative force field having potential energy function $U = - \left(\frac{1}{100} \text{ J/m}^4 \right) [x - 10]^2 [x - 30]^2$ Where x is in meters. Then work done by conservative force during the displacement from N to S is	(R)	+ 150 J
(IV)	A solid semi-sphere of mass 4 kg and radius 10 m is slowly turned as shown in the figure. Then work done by external force in turning is (take $g = 10 \text{ m/s}^2$) 	(S)	-150 J
		(T)	Zero

(A) I $\rightarrow$ R; II $\rightarrow$ T; III $\rightarrow$ P; IV $\rightarrow$ P

(B) I $\rightarrow$ S; II $\rightarrow$ T; III $\rightarrow$ Q; IV $\rightarrow$ P

(C) I $\rightarrow$ R; II $\rightarrow$ Q; III $\rightarrow$ P; IV $\rightarrow$ P

(D) I $\rightarrow$ T; II $\rightarrow$ Q; III $\rightarrow$ Q; IV $\rightarrow$ P

Space for Rough Work

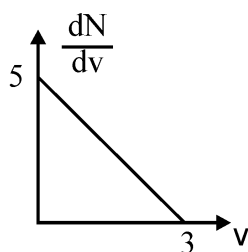
SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

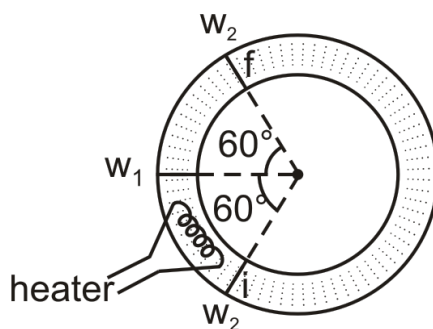
1. Consider a gas kept in a closed container. N is the number of molecules present in the container and v is the speed of molecules. Molecular speed distribution graph is shown in figure with suitable SI units of $\frac{dN}{dv}$ and v . Find v^2_{rms} of molecules in SI units.



Space for Rough Work

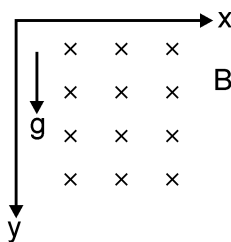
2. An insulated thin tube is fixed on horizontal surface. The total volume of tube is V_0 . Inside the tube, w_1 is fixed adiabatic wall and w_2 is movable adiabatic wall. An ideal gas of adiabatic coefficient $\gamma = 3/2$ is filled to entire volume of tube in both the chambers. The initial pressure in both the chambers is P_0 and number of moles of gas in both the chambers are same. A heater is fixed which supplies heat very slowly such that wall w_2 is moved from position i to f, at this position wall w_2 stays in equilibrium position even the heater is switched off. Neglect friction and any loss of heat to the surrounding. The total heat supplied by heater is $\mu P_0 V_0$. Find the value of μ .

($\sqrt{5} = 2.23$)



Space for Rough Work

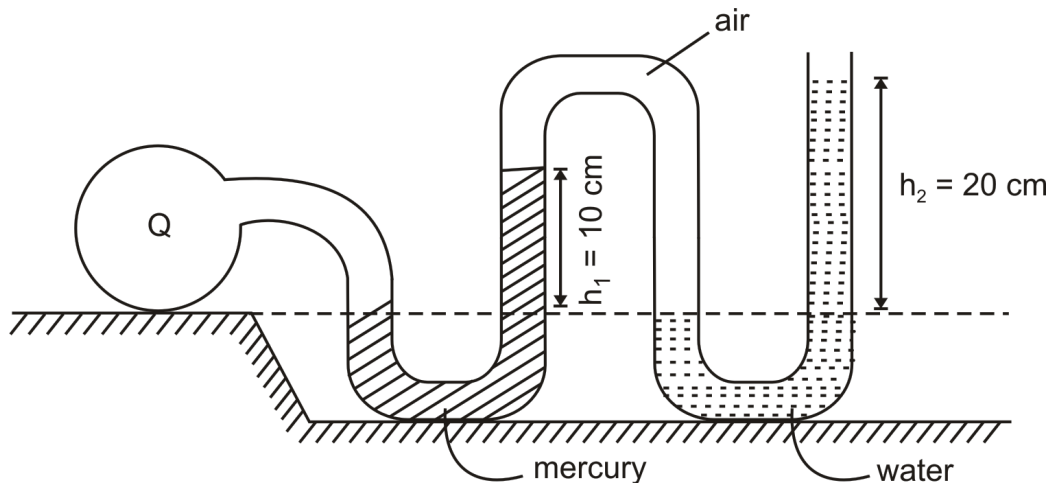
3. A constant uniform magnetic field $B = 5 \times 10^6$ T exist in a uniform vertical gravitational field $g = 10$ m/s². Both fields are perpendicular to each other. A particle of mass $m = 50$ gm and charge $q = 2\mu\text{C}$ is released in the space. Find the displacement (in cm) of particle along y-axis when velocity of particle is along x-axis first time.



4. In screw gauge, the graduation in the circular scale consists of 100 equal divisions and 10 complete rotation of the cap is equivalent to 5 mm on the main scale. While measuring diameter of a wire with screw gauge, a particular observation gives initial circular cap reading as 57 and after completing 3-complete rotation the final circular cap reading as 63. It is observed that while proceeding from initial reading to final reading, the cap reading increases. Calculate the diameter of wire (in mm) from the above data.

Space for Rough Work

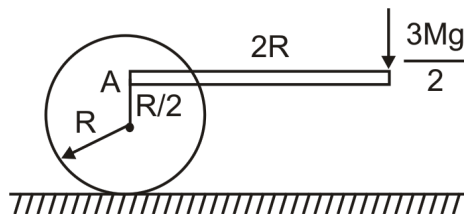
5. A bulb Q is filled with air. A tube is connected to the bulb filled with mercury & water as shown in figure. The heights of mercury and water are shown in the figure. If the pressure inside the bulb $n \times 10^5 \text{ N/m}^2$. Find the value of n if one end of tube opens to atmosphere and density of air to be small.
(take $g = 10 \text{ m/s}^2$, atmospheric pressure 10^5 N/m^2 , density of mercury = 13.8 gm/cm^3 and density of water = 1 gm/cm^3)



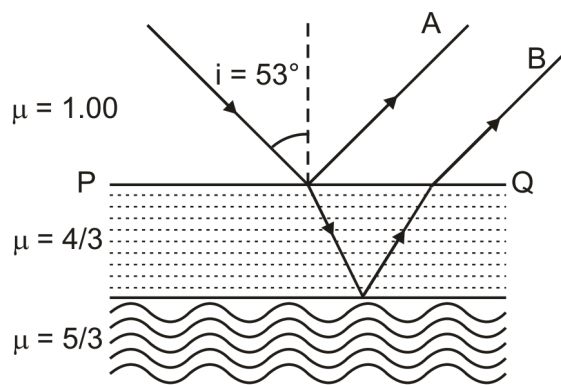
6. A man is looking down a 9 m deep tank filled with transparent liquid of refractive index $\sqrt{3}$, the line of sight of the man makes an angle 30° with the horizontal. What is the apparent depth of tank (in meter) as seen by the man ?

Space for Rough Work

7. A disc of mass M & radius R is placed on a rough horizontal surface. A light rod of length $2R$ is fixed to the disc at point A which is above centre and at a distance $R/2$ from centre as shown in figure. A force $\frac{3}{2} Mg$ is applied at the other end perpendicular to the rod. Find the minimum value of friction coefficient between the disc & ground if disc starts rolling without sliding.



8. A light wave of wavelength $\lambda = 650$ nm is incident on the surface PQ at an angle of incidence $i = 53^\circ$ as shown in figure. If the phase difference between the two reflected rays A & B is π . Find thickness (in nano-meter) of the film of refractive index $4/3$.



Space for Rough Work

PART-2 : CHEMISTRY
SECTION-I (i) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

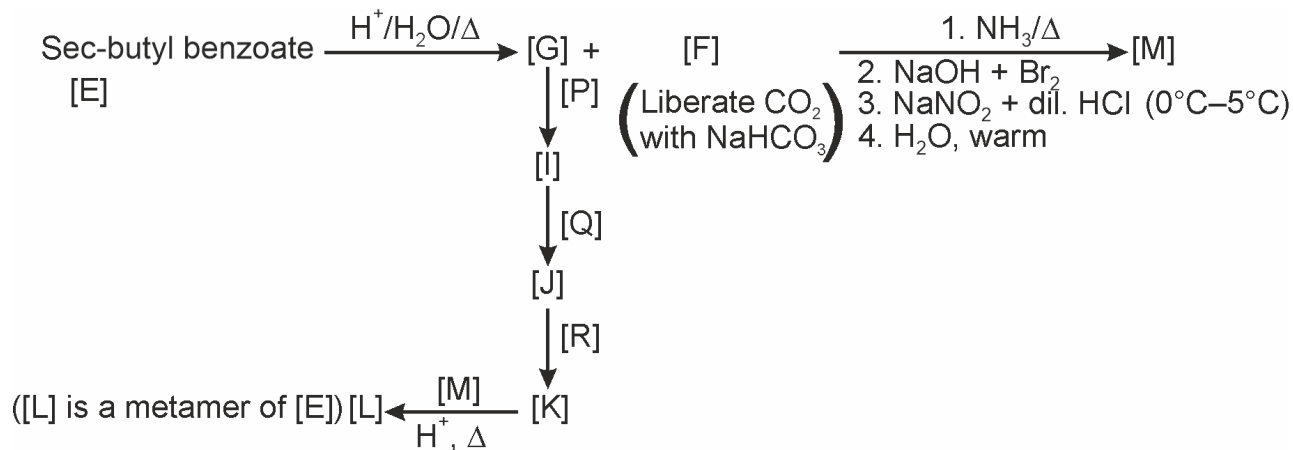
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

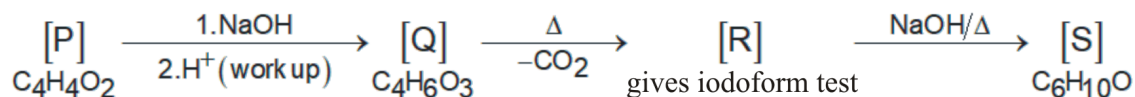
1. Choose the correct options regarding the given scheme?



- (A) [P] = $SOCl_2$, [Q] = Mg/dry ether, [R] = 1. HCHO 2. H^+
- (B) [P] = $KBr + H_2SO_4$, [Q] = AgCN/DMF, [R] = H_3O^+/Δ
- (C) [P] = PBr_3 , [Q] = Mg / dry ether, [R] = 1. CO_2 2. H^+
- (D) [P] = HBr, [Q] = NaCN/DMF, [R] = H_3O^+/Δ

Space for Rough Work

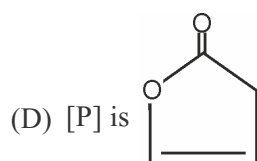
2. Choose the INCORRECT statement(s) for the given scheme



Unsaturated

Lactone

- (A) [S] is an α, β -unsaturated carbonyl compound.
 (B) Four stereoisomers are possible for [S]
 (C) Two structural isomers of [P] can satisfy the above reaction scheme.



3. Which of the following order is correct according to indicated property ?

- (A) $\text{SiO}_2 > \text{LiF} > \text{H}_2\text{O} > \text{CCl}_4 > \text{H}_2$ (Melting point)
 (B) $\text{BeCO}_3 > \text{SrCO}_3 > \text{BaCO}_3$ (K_p value at same temperature)
 (C) $\text{RNH}_2 > \text{R}_2\text{NH} > \text{R}_3\text{N}$ (Boiling point)
 (D) $\text{NO}^{2-} > \text{NO}^- > \text{NO} = \text{NO}^{2+} > \text{NO}^+$ (Bond length)

Space for Rough Work

4. Select the correct statements
- (A) Compound Sodium hexameta phosphate contains six P–O–P linkage and 24 sigma bonds.
- (B) Na_2CO_3 is prepared by solvay's process but K_2CO_3 can not be prepared because K_2CO_3 is highly water soluble.
- (C) BeCl_2 can exist as linear molecule in the vapour state and it exists as polymer in solid state.
- (D) In solid state dihedral angle of H_2O_2 is different from its gaseous state.
5. Which of the following statements is/are correct :-
- (A) Conductivity decreases with decrease in concentration for strong as well as weak electrolytes
- (B) SO_4^{2-} will be oxidised in the electrolysis of aqueous Na_2SO_4 using Hg as cathode and Pt as anode.
- (C) Conductance(G) depends on cell constant as well as concentration
- (D) The limiting equivalent conductivity of weak electrolyte can't be determined exactly by extra plot of the plot of Λ_{eq} against $\sqrt{c}$.
6. Which of the following statements is/are incorrect :-
- (A) A catalyst can cause a non spontaneous reaction to proceed
- (B) A catalyst decreases $\Delta_r H$ of reaction.
- (C) Order of reaction may be zero, integral or a fraction but molecularity is always a non-zero whole number and never a fraction.
- (D) If the partial orders differ from the stoichiometric co-efficients in the balanced reaction, then it must be complex reaction.

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each **question has matching lists**. The codes for the lists have choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If ONLY the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

7. Match each species in List-I with appropriate statement(s) in List-II and choose correct options:

List-I		List-II	
(I)		(P)	More than 4 moles of HI will be consumed.
(II)		(Q)	Product react with Na/dry ether.
(III)		(R)	Product formed contains 2° halide.
(IV)		(S)	CH ₃ I is one of the product
		(T)	2-Iodopropane is a major product

- (A) I → Q; II → P, Q, R; III → P, Q, R; IV → Q, S, T
- (B) I → P, T; II → Q, R, S; III → P, Q, R, T; IV → P, Q, R
- (C) I → P, Q; II → Q, S, T; III → P, Q, R; IV → Q, R, T
- (D) I → Q, S; II → P, Q, R, S; III → P, Q, R; IV → P, Q, R, T

Space for Rough Work

8. Match each species in List-I with List - II and choose correct option

List-I		List-II	
(I)	A carbohydrate which yields only glucose on hydrolysis	(P)	Sucrose
(II)	A carbohydrate which yields glucose and fructose on hydrolysis	(Q)	Lactose
(III)	A carbohydrate which yields glucose and galactose on hydrolysis	(R)	Amylose
(IV)	A carbohydrate which can reduce Fehling's solution	(S)	Mannose
		(T)	Cellulose

(A) I $\rightarrow$ R,T;II $\rightarrow$ Q;III $\rightarrow$ P;IV $\rightarrow$ Q,S

(B) I $\rightarrow$ T;II $\rightarrow$ Q;III $\rightarrow$ Q;IV $\rightarrow$ Q

(C) I $\rightarrow$ R;II $\rightarrow$ P;III $\rightarrow$ Q;IV $\rightarrow$ P

(D) I $\rightarrow$ R,T;II $\rightarrow$ P;III $\rightarrow$ Q;IV $\rightarrow$ Q,S

9. Match the list

List-I		List-II	
(I)	F, Cl, Br, I	(P)	Increasing order of atomic radius
(II)	N, F, O, Li	(Q)	Decreasing order of electronegativity
(III)	N, B,O, Cl	(R)	Corresponding elements will lie on similar positions in Lothar Mayer's curve.
(IV)	F, O, N, C	(S)	Increasing order of EA
		(T)	Increasing order of 2 nd I.E.

(A) I $\rightarrow$ R,P,Q;II $\rightarrow$ T;III $\rightarrow$ S;IV $\rightarrow$ Q

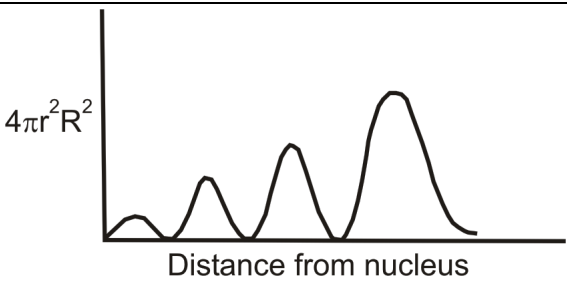
(B) I $\rightarrow$ R,P,Q;II $\rightarrow$ T,Q;III $\rightarrow$ S,P;IV $\rightarrow$ P

(C) I $\rightarrow$ R,P,Q;II $\rightarrow$ T;III $\rightarrow$ S;IV $\rightarrow$ Q,P

(D) I $\rightarrow$ R,P;II $\rightarrow$ S;III $\rightarrow$ T;IV $\rightarrow$ P,Q

Space for Rough Work

10. Match the list

List-I		List-II	
(I)	 Distance from nucleus	(P)	5p _y
(II)	Angular probability depends upon θ and ϕ	(Q)	8
(III)	If $A = \frac{2KE_{2,3}}{3PE_{4,2}}$ then value of A is $E_{n,z}$ = energy of electron in n^{th} orbit, $z \rightarrow$ atomic number.	(R)	3
(IV)	Sum of number of maxima in the curve $4\pi r^2 R^2$ vs r for 4p _y and 7d _{xy} .	(S)	4p _y
		(T)	6d _{xy}

(A) I $\rightarrow$ P,T;II $\rightarrow$ P,T,S;III $\rightarrow$ R;IV $\rightarrow$ Q

(B) I $\rightarrow$ P,T;II $\rightarrow$ P,T,S;III $\rightarrow$ Q;IV $\rightarrow$ R

(C) I $\rightarrow$ T;II $\rightarrow$ P,S;III $\rightarrow$ R;IV $\rightarrow$ Q

(D) I $\rightarrow$ T;II $\rightarrow$ P,S;III $\rightarrow$ Q;IV $\rightarrow$ R

Space for Rough Work

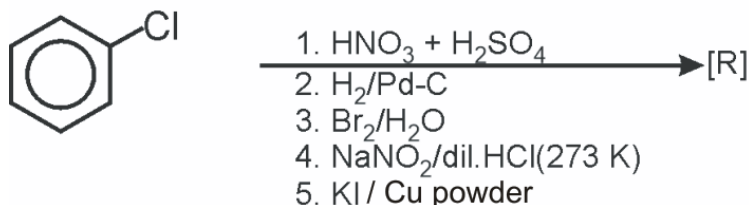
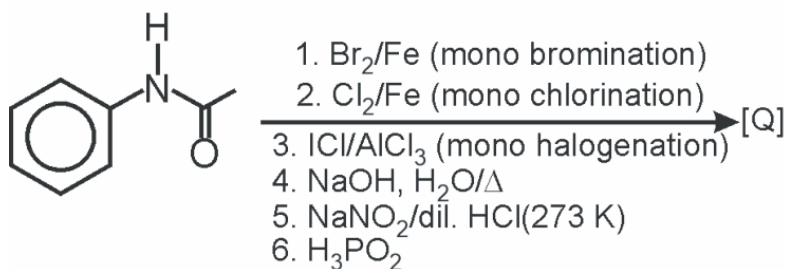
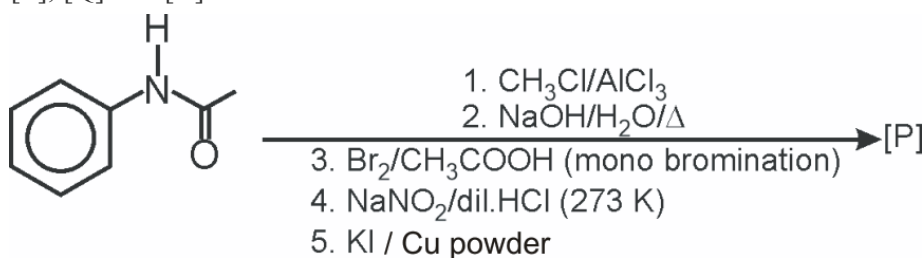
SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered.

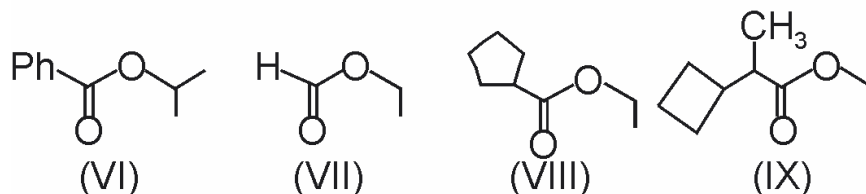
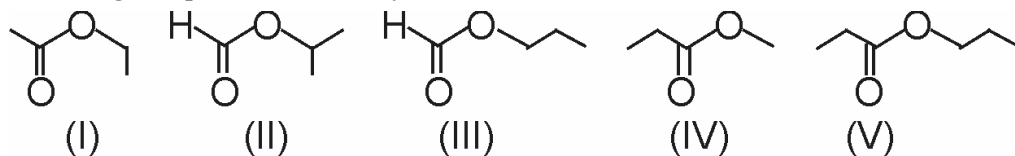
Zero Marks : 0 In all other cases.

1. Consider the major organic product in each case and calculate the sum of number of halogen atoms in [P], [Q] and [R]



Space for Rough Work

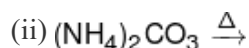
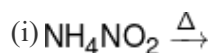
2. An organic compound reacts with excess of MeMgBr followed by hydrolysis to give two organic products of which one reacts with NaOI to give yellow precipitate and OTHER DOES NOT. How many of the following compounds can satisfy these conditions?



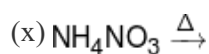
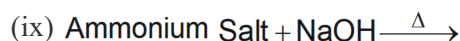
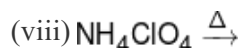
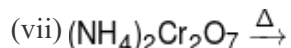
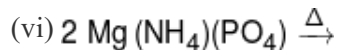
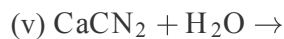
3. Consider all possible isomeric ketones including stereoisomers of $\text{MW} = 100$. All these isomers are independently reacted with CH_3MgBr . The total number of ketones (including stereoisomers) that gives a racemic product is/are

Space for Rough Work

4. Consider the following reaction :-



(iii) Heating of microcosmic salt at high temperature

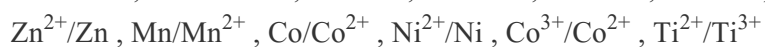
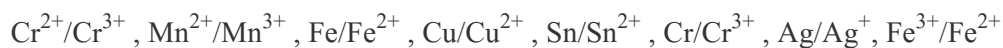


Number of reactions producing Ammonia gas = P

Number of reactions producing Nitrogen gas = R

Write your answer as [P×R]

5. If number of process having positive value of standard electrode potential and negative value of standard electrode potential are X and Y respectively then what is the value of [X ÷ Y] ?



Space for Rough Work

6. 4 g of rock salt type ionic solid(AB) is dissolved in water to make 4 L solution. It is salt of strong base and weak acid whose pH is 9. If distance between cation and anion is 400 pm. Calculate density of ionic solid(AB) in g/ml.
($pK_w = 14$, $pK_a(HA) = 6$, $N_A = 6 \times 10^{23}$).
7. A one litre reaction vessel which is equipped with a movable piston is filled completely with a 8 M aqueous solution of H_2O_2 . Decomposition of H_2O_2 is a first order reaction with half life 4 hours at 300 K. Due to formation of gas, piston moves against a constant external pressure of 2 atm. Calculate the magnitude of work done (in KJ) by the gas from the start of 5th hour till the end of 12th hours. ($R = 8.3 \text{ J/mol-K}$)
8. When 1.4 g oxalic acid is burnt in bomb calorimeter at 27°C, the temperature of the calorimeter is increased by 2.5°C. If $\Delta H^\circ_{\text{combustion}}$ for oxalic acid(s) is -300 kJ/mol at 27°C, then calculate the heat capacity of the calorimeter_____.(in kJ/K). [Given : $R = 8.3 \text{ J/K-mol}$]

Space for Rough Work

PART-3 : MATHEMATICS
SECTION-I (i) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. Let $P(x)$ be a polynomial of least degree with leading coefficient 1 and rational coefficients, such that

$$P\left(\sqrt{5 + \sqrt{5 + \sqrt{5 + \dots \infty \text{ terms}}}}\right) = 0, \text{ then :-}$$

(A) The value of $100 \sum_{r=2}^{100} \frac{1}{P(r) + 5}$ is equal to 99

(B) $\lim_{n \rightarrow \infty} \sum_{r=3}^n \frac{1}{P(r) + 3}$ is equal to 1

(C) $\int_0^1 \frac{dx}{P(x) + x + 6}$ is equal to $\frac{\pi}{4}$

(D) $\int_1^2 \frac{dx}{\sqrt{P(x) + x + 4}}$ is equal to $\ln(2 + \sqrt{3})$

Space for Rough Work

2. For $x \in \left(0, \frac{\pi}{4}\right)$

$$R_n = \sum_{r=1}^{2n} \sin(\sin^{-1} x^{4r-3}),$$

$$S_n = \sum_{r=1}^{2n} \cos(\cos^{-1} x^{4r-2}),$$

$$T_n = \sum_{r=1}^{2n} \tan(\tan^{-1} x^{4r-1}),$$

$$U_n = \sum_{r=1}^{2n} \cot(\cot^{-1} x^{4r})$$

where $n \in \mathbb{N}$ and $n \geq 4$:

(A) $R_n < S_n < T_n < U_n$

(B) $R_n > S_n > T_n > U_n$

(C) $\lim_{n \rightarrow \infty} (R_n + S_n + T_n + U_n)$ is equal to $\frac{x}{1-x}$

(D) The value of x for which $R_n + S_n = T_n + U_n$ is $2\sin \frac{\pi}{10}$

Space for Rough Work

3. Which of the following statement(s) is/are correct?
- (A) The number of quadratic equations having real roots which remain unchanged even after squaring their roots is 3.
- (B) The number of solutions of the equation $\tan 2\theta + \tan 3\theta = 0$, in the interval $[0, \pi]$ is equal to 6.
- (C) For $x_1, x_2, x_3 > 0$, the minimum value of $\frac{2x_1}{x_2} + \frac{128x_3^2}{x_2^2} + \frac{x_2^3}{4x_1x_3^2}$ is equal to 24.
- (D) The locus of the mid-points of chords of the circle $x^2 + y^2 - 2x - 6y - 1 = 0$, which are passing through origin is $x^2 + y^2 - x - 3y = 0$.
4. Let A, B, C in order roll a fair dice indefinitely untill a winner is obtained. A wins if he rolls a prime number & B wins if he rolls a composite number, whereas C wins if he rolls a neither prime nor composite number. Let $P(A)$, $P(B)$ and $P(C)$ denote their respective chances of winning, then identify the correct statement(s).
- (A) $P(A) - P(B) = \frac{6}{13}$
- (B) $P(B) - P(C) = \frac{4}{13}$
- (C) $P(C) + P(A) = \frac{10}{13}$
- (D) $P(A) = 3P(B)$

Space for Rough Work

5. Let $\vec{p}$, $\vec{q}$, $\vec{r}$ are non-zero vectors such that $\vec{r} \times (\vec{p} \times \vec{q}) = \vec{p} \times (\vec{p} \times \vec{q})$. If $|\vec{p} \times \vec{q}| = 3$, $[\vec{p} \vec{q} \vec{r}] = 12$ and $\vec{p} \cdot \vec{p} = 4$, then :-

(A) $|\vec{r}| = 2\sqrt{10}$

(B) $|\vec{r} - \vec{p}| = 4$

(C) $(\vec{r} - \vec{p}) \cdot \vec{r} = 16$

(D) $|\vec{r} - \vec{p}| = 6$

6. If $I_1 = \frac{\int_0^{\frac{\pi}{4}} (t - \frac{\pi}{4})^{100} \sec^2 t dt}{\int_0^{\frac{\pi}{4}} (t - \frac{\pi}{4})^{99} \tan t dt}$ and $I_2 = \frac{1}{100} \int_0^{\frac{\pi}{4}} (t - \frac{\pi}{4})^{100} \sec^2 t dt - \int_0^{\frac{\pi}{4}} (t)^{99} \left(\frac{1 - \tan t}{1 + \tan t} \right) dt$, then :-

(A) $I_1 < I_2$

(B) $I_1 > I_2$

(C) $I_1 I_2 = 0$

(D) $I_1 I_2 = 9900$

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each **question has matching lists**. The codes for the lists have choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If ONLY the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

7. If an equilateral triangle ABC with vertices at z_1, z_2 and z_3 be inscribed in a circle $|z| = 2$ and again a circle is inscribed in $\triangle ABC$ touching sides AB, BC and CA at $D(z_4), E(z_5)$ and $F(z_6)$ respectively.

List-I		List-II	
(I)	Value of $\operatorname{Re}(z_1 \bar{z}_2 + z_2 \bar{z}_3 + z_3 \bar{z}_1)$ is equal to	(P)	2
(II)	If $\frac{4z_1}{z_3} = a(-1 + i\sqrt{3})$ then a is equal to	(Q)	-6
(III)	$ z_1 + z_2 ^2 + z_2 + z_3 ^2 + z_3 + z_1 ^2$ is equal to	(R)	12
(IV)	If P is any point on incircle, then $DP^2 + EP^2 + FP^2$ is	(S)	-12
		(T)	6

(A) $I \rightarrow Q; II \rightarrow R; III \rightarrow T; IV \rightarrow R$

(B) $I \rightarrow R; II \rightarrow S; III \rightarrow P; IV \rightarrow Q$

(C) $I \rightarrow Q; II \rightarrow P; III \rightarrow R; IV \rightarrow T$

(D) $I \rightarrow T; II \rightarrow R; III \rightarrow Q; IV \rightarrow P$

Space for Rough Work

8.

List - I		List - II	
(I)	If $\int_0^a \ln(\cot a + \tan x) dx$, where $a \in \left(0, \frac{\pi}{2}\right)$, is equal to $\lambda a \ln \operatorname{cosec}(a)$, ($\lambda \in \mathbb{R}$) then the value of λ is equal to	(P)	0
(II)	If coefficient of x^{2017} in $(1 + x + x^2 + x^3 + x^4)^{1001} \cdot (1 - x)^{1002}$ is $k \cdot {}^{1001}C_{503}$, where $k \in \mathbb{R}$, then $ k $ is equal to	(Q)	1
(III)	A hyperbola passes through the point on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, ($a > b$) whose eccentric angle is $\frac{\pi}{4}$. If the hyperbola and ellipse have the same foci and eccentricity of hyperbola is e , then e^2 is :-	(R)	2
(IV)	If the fractional part of $\frac{1}{x}$ and x^2 for some $x \in (\sqrt{2}, \sqrt{3})$ are equal then the value of $x^4 - \frac{3}{x}$ is	(S)	3
		(T)	5

(A) $I \rightarrow P; II \rightarrow Q; III \rightarrow T; IV \rightarrow S$

(B) $I \rightarrow Q; II \rightarrow R; III \rightarrow S; IV \rightarrow P$

(C) $I \rightarrow T; II \rightarrow S; III \rightarrow P; IV \rightarrow Q$

(D) $I \rightarrow Q; II \rightarrow P; III \rightarrow R; IV \rightarrow T$

Space for Rough Work

9.

List - I		List - II	
(I)	Let S_1 and S_2 are two points on AB of a ΔABC with vertices A $(-2, 3)$, B $(4, -6)$ and C $(1, 1)$. CS_1 and CS_2 divide the triangle into three parts of equal area. If the equation of the lines through the origin drawn parallel to CS_1 and CS_2 is $y^2 + \lambda xy - \mu x^2 = 0$, then the value of $(\lambda + \mu)$ is :-	(P)	2
(II)	If the area enclosed by the curves $y = x $ and $x^2 + y - 2 = 0$ is $\frac{p}{q}$, where p and q are coprime number, then the value of $(p - q)$ is :-	(Q)	4
(III)	The curve that satisfies the differential equation $\frac{dy}{dx} = \frac{x^2 + y^2}{2xy}$ is a hyperbola passing through $(2, 1)$ with eccentricity e. The value of e^4 is equal to	(R)	6
(IV)	$\lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{\ln(n^2 + r^2) - 2n \ln(n)}{n} \text{ Let } = \ln 2 + \frac{\pi}{2} - 2,$ If $\lim_{n \rightarrow \infty} \frac{1}{n^{2m}} \left[(n^2 + 1^2)^m (n^2 + 2^2)^m \dots (n^2 + n^2)^m \right]^{\frac{1}{n}} = \left(\frac{a \sqrt{e\pi}}{e^2} \right)^m$, then the value of a is :-	(S)	7
		(T)	1

(A) I $\rightarrow$ S; II $\rightarrow$ Q; III $\rightarrow$ R; IV $\rightarrow$ T

(B) I $\rightarrow$ S; II $\rightarrow$ Q; III $\rightarrow$ Q; IV $\rightarrow$ P

(C) I $\rightarrow$ R; II $\rightarrow$ S; III $\rightarrow$ P; IV $\rightarrow$ Q

(D) I $\rightarrow$ Q; II $\rightarrow$ P; III $\rightarrow$ T; IV $\rightarrow$ S

Space for Rough Work

10.

List - I		List - II	
(I)	If the distance of the plane passing through the point P(1, 1, 1) and perpendicular to the line $\frac{x-1}{3} = \frac{y-1}{0} = \frac{z-1}{4}$ from the origin is $\frac{p}{q}$ (where p and q are co-prime number), then the value of (p - q) is	(P)	1
(II)	The interval [0, 4] is divided into n equal sub-intervals by the points $x_0, x_1, x_2, \dots, x_{n-1}, x_n$, where $0 = x_0 < x_1 < x_2 < x_3 < \dots < x_n = 4$. If $\delta x = x_i - x_{i-1}$, for $i = 1, 2, 3, \dots, n$, then the value of $\lim_{\delta x \rightarrow 0} \sum_{i=1}^n x_i \delta x$ is	(Q)	2
(III)	Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function satisfying $x f(x) + (1-x) f(-x) = x^2 + x + 1$ for any real number x. The greatest real number M for which $f(x) \geq M$ for all real numbers x, is equal to $\frac{p}{q}$, (where p and q are co-prime number), then the value of (q - p) is :-	(R)	3
(IV)	The area of the trapezium ABCD with $AB \parallel CD$, $AD \perp AB$ and $AB = 3CD$ is equal to 4. A circle inside the trapezium is tangent to all of its sides. If the radius of the circle is r, then the value of $4r^2$ is :-	(S)	6
		(T)	8

(A) I $\rightarrow$ P; II $\rightarrow$ Q; III $\rightarrow$ R; IV $\rightarrow$ T

(B) I $\rightarrow$ S; II $\rightarrow$ Q; III $\rightarrow$ T; IV $\rightarrow$ P

(C) I $\rightarrow$ Q; II $\rightarrow$ T; III $\rightarrow$ P; IV $\rightarrow$ R

(D) I $\rightarrow$ Q; II $\rightarrow$ P; III $\rightarrow$ T; IV $\rightarrow$ S

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

1. If the number of triplets (a, b, c) , where $a, b, c \in \mathbb{N}$ such that origin and the point $(1, 1, 4)$ are lying on the same side of the plane $ax + by + cz = 41$ is M , then find the value of $\left(\frac{M}{730}\right)$.
2. Let a, b and c be three distinct real numbers in G.P. such that $a < 0, b > 0$ and $c < 0$. If A.M. of b and c be a and r is the common ratio of G.P., then find the value of $\int_{r/2}^{-r/2} (x^5 + 3x^2 + \text{sgn}([x]) + [x] + 1) dx$.
(where $[k]$ denotes greatest integer less than or equal to k)

Space for Rough Work

3. Let $f(x)$ be a polynomial satisfies the relation $f(f(f(x))) + (1-p)f(x) = 3, \forall x \in \mathbb{R}$, where p is any real number. If leading coefficient of $f(x)$ is 2, then find the value of

$$\left(\frac{d}{dx}(f(f(x))) \text{ at } x = p \right) + \int_{-p}^p (2f^{-1}(x) + 1) dx.$$

4. A point $P(\sqrt{2} \sin A, \cot B)$ lies on the curve $(x+y)^2 + 4(y-x)^2 = 8$. If minimum value of $(2 \tan C - \sqrt{2} \sin A)^2 + (2 \cot C - \cot B)^2 = a + b\sqrt{2}$, where $C \in \mathbb{R} - \left\{ \frac{n\pi}{2} \right\}; n, a, b \in \mathbb{I}$, then find the value of $(a+b)$.

5. A 3×3 determinant has entries either 1 or -1. Let S be the set of all determinants such that the product of elements of any row or column is -1. For example $\begin{vmatrix} 1 & -1 & 1 \\ 1 & 1 & -1 \\ -1 & 1 & 1 \end{vmatrix}$ is an element of S and number of elements in S is m . Let $P = \begin{bmatrix} 3 & -2 & 3 \\ 2 & -2 & 3 \\ 0 & -1 & 1 \end{bmatrix}$ and trace of the matrix $\text{adj}(\text{adj } P)$ is n , then find the value of $\frac{m}{n}$.

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6. Given two intersecting circles C_1 and C_2 with centres at M and N respectively. From one of the point of intersection A of circles, tangents are drawn to two circles meeting the circles at B and C respectively. Let P be any point located such that PMAN is a parallelogram and if $AB + BC + CA = K(PA \sin A + PB \sin B + PC \sin C)$, $k \in \mathbb{R}$ then find k.
(where angles A, B, C are angles of $\triangle ABC$)
7. If $x = \log_2 3 - 4 \log_{\left(\ln \frac{5}{4}\right)} 3$, $y = \log_{\sqrt{3}} \left(\ln \frac{5}{4}\right)$ and $z = \log_2 \left(\log_{(5/4)} e\right)$, then find the absolute value of $(xy + 2z)$.
8. If $3 \tan^6 10^\circ - 27 \tan^4 10^\circ + 33 \tan^2 10^\circ < \log_2 [x - 2]$, then find the minimum value of x :-
(where $[k]$ denotes greatest integer less than or equal to k)

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NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them._____
Signature of the CandidateI have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same._____
Signature of the Invigilator

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